

Machine Learning 1

Lecture 2

Estimation

Alejandro Monroy

Learning goals

- Define what an estimator is
- Compute the bias of an estimator
- Understand and apply the Maximum Likelihood Estimation (MLE) principle to compute parameter estimates
- Formulate Maximum A Posteriori (MAP) estimation by applying Bayes' theorem to incorporate parameter priors directly into the optimization objective
- Similarly, derive the probability distribution of the posterior

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Outline

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Lecture 2

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Last lecture

Probability distributions

- **PDF:** $\mathbb{P}[a \leq X \leq b] = \int_a^b p(x) dx$
- **CDF:** $P(x) = \mathbb{P}[X \leq x] = \int_{-\infty}^x p(t) dt$

Probability formulas

- **Sum rule:** $p(x) = \int_y p(x, y) dy$
- **Product rule:** $p(x, y) = p(x)p(y|x)$
- **Bayes rule:** $p(y | x) = \frac{p(x|y)p(y)}{p(x)}$

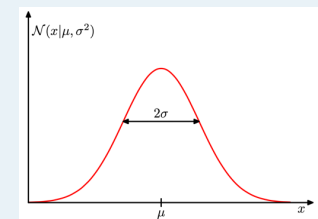
Moments

- $\mathbb{E}[X] = \int_{-\infty}^{\infty} x p(x) dx$
- $\text{var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$
- $\text{cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$

Gaussian distribution

$$\mathcal{N}(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

- $\mathbb{E}[X] = \mu$
- $\text{var}[X] = \sigma^2$



Estimation

Estimation

Goal: Given a model $p(x)$, we want to compute an estimate of some quantity of interest based on observed data $\mathcal{D} = \{x_1, \dots, x_N\}$.

Examples:

- The sample mean is an estimator of the population mean $\mathbb{E}[x]$

$$\bar{x} = \frac{\sum_i x_i}{N}$$

- The sample variance is an estimator of the population variance

$$\tilde{\sigma}^2 = \frac{\sum_i (x_i - \bar{x})^2}{N}$$

Estimation - Bias

The **bias** of an estimator is the difference between its expected value and the true (population) value of the parameter:

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

We say that an estimator $\hat{\theta}$ is **unbiased** iff $\text{Bias}(\hat{\theta}) = 0$

Are the estimators from the previous slide unbiased?

$$x \sim p(x) \\ x_i \sim p(x) \forall i$$

- Sample mean:

$$\mathbb{E}[\bar{x}] = \mathbb{E}\left[\frac{\sum_{i=1}^N x_i}{N}\right] = \frac{\sum_{i=1}^N \mathbb{E}[x_i]}{N} = \frac{\sum_{i=1}^N \mathbb{E}[x]}{N} = \frac{N \mathbb{E}[x]}{N} = \mathbb{E}[x]$$

$\Rightarrow \text{Bias}(\bar{x}) = 0 \Rightarrow \bar{x}$ is an unbiased estimator of $\mathbb{E}[x]$

Estimation - Bias

Are the estimators from the previous slide unbiased? We assume that the samples are i.i.d.

• Sample variance: $E[\hat{S}^2] = E\left[\frac{\sum (x_i - \bar{x})^2}{N}\right]$

(linearity of E)

$$= \frac{1}{N} \sum_i E[x_i^2 - 2x_i\bar{x} + \bar{x}^2]$$

$$= \frac{1}{N} \sum_i \left(E[x_i^2] - 2E\left[x_i \frac{\sum_j x_j}{N}\right] + \frac{1}{N^2} E\left[\left(\sum_j x_j\right)^2\right] \right)$$

$$= \frac{1}{N} \sum_i \left(E[x_i^2] - \frac{2}{N} \left(\sum_{j \neq i} E[x_i x_j] + E[x_i^2] \right) + \frac{1}{N^2} \sum_j \left(\sum_{k \neq j} E[x_j x_k] + E[x_j^2] \right) \right)$$

$$\stackrel{*}{=} \frac{1}{N} \sum_i \left(E[x^2] - \frac{2}{N} \left(\underbrace{\sum_{j \neq i} E[x_i] E[x_j]}_{E[x^2]} + E[x^2] \right) + \frac{1}{N^2} N(N-1) E[x]^2 + E[x^2] \right) \quad (x_i \perp x_j \ \forall i \neq j)$$

* Note: If x_i and x have the same distribution, then: $E[x_i] = E[x]$
 $E[x_i^2] = E[x^2]$

Estimation - Bias

$$= \frac{1}{N} \sum_i \left(\mathbb{E}[x^2] - \frac{2}{N} \left(\underbrace{\sum_{j \neq i} \mathbb{E}[x_i] \mathbb{E}[x_j]}_{\mathbb{E}[x]^2} + \mathbb{E}[x^2] \right) + \frac{1}{N^2} (N-1) \mathbb{E}[x]^2 + \mathbb{E}[x^2] \right) \quad (\text{previous slide})$$

$$= \frac{1}{N} \sum_i \left(\mathbb{E}[x^2] - \frac{2}{N} (N-1) \mathbb{E}[x]^2 - \frac{2}{N} \mathbb{E}[x^2] + \frac{N-1}{N} \mathbb{E}[x]^2 + \frac{1}{N} \mathbb{E}[x^2] \right)$$

$$= \frac{1}{N} \sum_i \left(\frac{N-2+1}{N} \mathbb{E}[x^2] + \frac{-2N+2+N-1}{N} \mathbb{E}[x]^2 \right)$$

$$= \frac{1}{N} \sum_i \left(\frac{N-1}{N} (\text{var}[x] + \cancel{\mathbb{E}[x]^2}) - \frac{N-1}{N} \cancel{\mathbb{E}[x]^2} \right) \quad (\text{var}[x] = \mathbb{E}[x^2] - \mathbb{E}[x]^2)$$

$$= \frac{1}{N} \cdot \cancel{N} \cdot \frac{N-1}{N} \text{var}[x]$$

$$\text{Bias}(\tilde{S}^2) = \mathbb{E}[\tilde{S}^2] - \text{var}(x) = \frac{N-1}{N} \text{var}[x] \neq 0 \Rightarrow \tilde{S}^2 \text{ is } \underline{\text{biased}}!$$

$$\text{Unbiased version} \rightarrow S^2 = \frac{N}{N-1} \tilde{S}_N^2 = \frac{\sum (x_i - \bar{x})^2}{N-1} \quad \left(\text{You can check yourself that } \mathbb{E}(S_N) = \text{var}[x] \right)$$

Parametric Estimation

Goal: Given a parametric model $p(x|\theta)$, we want to compute an estimate $\hat{\theta}$ of the parameter θ based on observed data $\mathcal{D} = \{x_1, x_2, \dots, x_N\}$.

For example, in a normally distributed random variable $x \sim \mathcal{N}(\mu, \sigma^2)$, the parameters are the population mean μ and the variance σ^2 , so we can use the estimators from the previous slide as estimators for this parameters

How to compute good parametric estimators?

In the case of the normal distribution, it is straight-forward to come up with good estimators, but how do we do it for distributions in which we don't know the "meaning" of the parameters?

We we will focus on two estimation methods:

- ***Maximum Likelihood Estimation (MLE)***
- ***Maximum a Posteriori Estimation (MAP)***

Maximum Likelihood Estimation

The Likelihood Function

Measures the plausibility of a model given a set of observed data

One sample:

$$\mathcal{L}(\theta) = p(x|\theta)$$

(It's just the probability density of the data, but viewed as a function of the parameters instead of the data, which is treated as fixed)

$$\mathcal{D} = \{x_1, \dots, x_N\}$$

N independent samples:

$$\mathcal{L}(\theta) = p(\mathcal{D}|\theta) = \prod_{i=1}^N p(x_i|\theta)$$

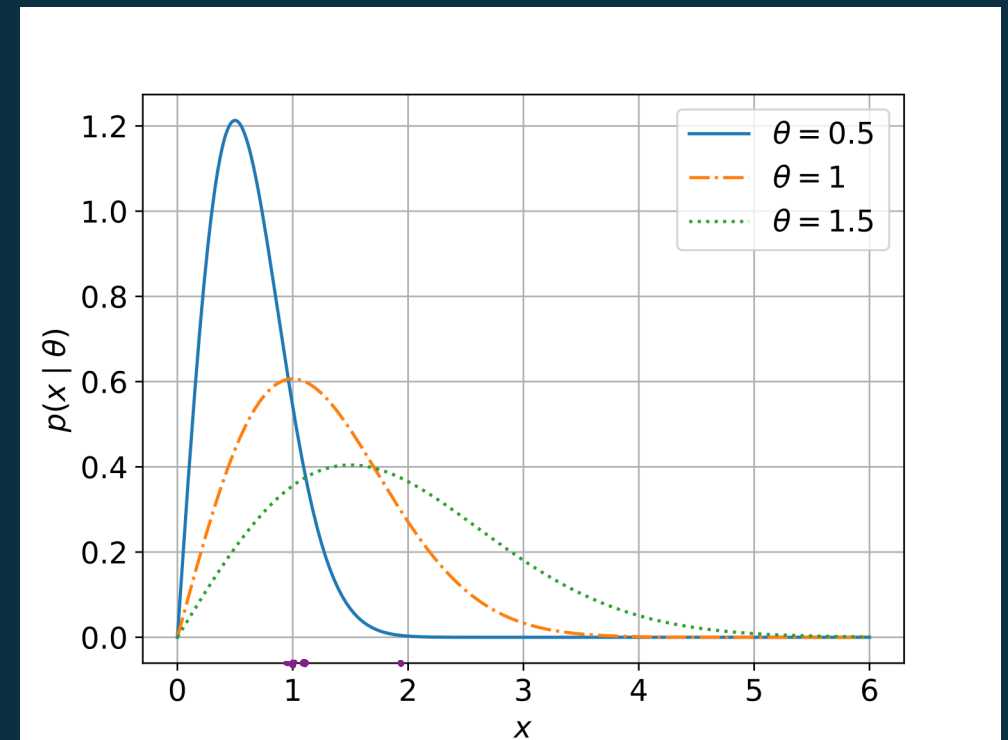
↑
indep. and
identically distributed (i.i.d.)

Quiz time!

We consider a random variable x modelled by a probability density function $p(x|\theta)$. The plot below shows the density function for three different values of θ .



Let $\mathbf{x} = \{0.95, 1.00, 1.15, 1.95\}$ be a set of independent samples of x . For which of the three values of θ shown in the plot ($\theta \in \{0.5, 1, 1.5\}$) is the likelihood $p(\mathbf{x}|\theta)$ maximal?



$$p(\underline{x}|\theta) \stackrel{\text{i.i.d}}{=} \prod_{i=1}^4 p(x_i|\theta)$$

- a) ~~$\theta = 0.5$~~
- b) $\theta = 1$
- c) $\theta = 1.5$
- d) $p(\mathbf{x}|\theta)$ has the same value for all three values of θ .

Maximum Likelihood Estimation

Setup: N independent samples, identically distributed according to $p(x | \theta)$, where θ is unknown.

Goal: find the value of the parameter θ that maximizes the likelihood

$$\hat{\theta}_{ML} = \underset{\theta}{\operatorname{argmax}} p(\mathcal{D} | \theta) = \underset{\theta}{\operatorname{argmax}} \prod_{i=1}^N p(x_i | \theta)$$

Maximum Likelihood Estimation – The log trick

Usually, we take the logarithm of the likelihood to improve numerical stability:

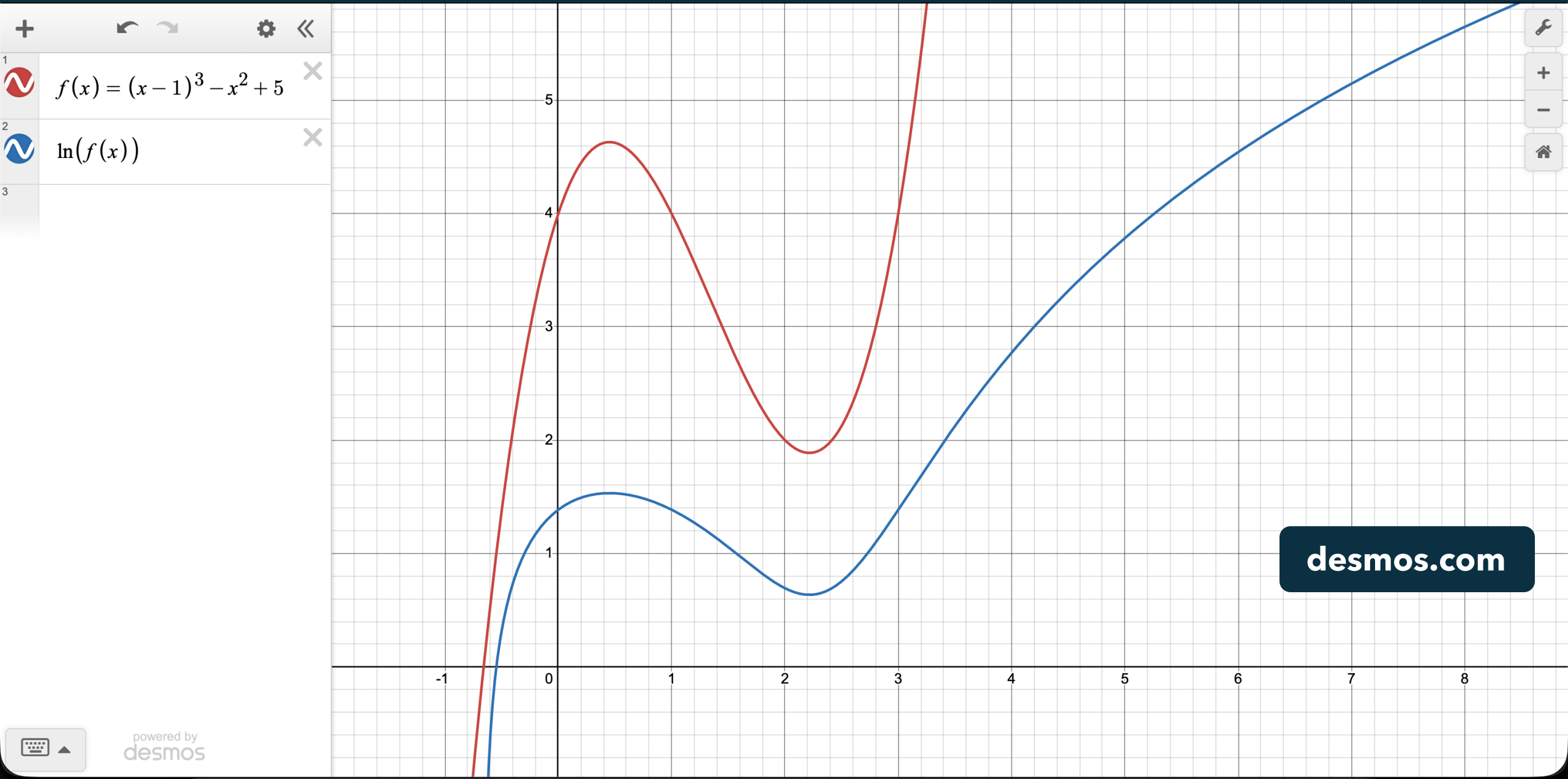
$$\theta_{ML} = \operatorname{argmax}_{\theta} \log \prod_{i=1}^N p(x_i | \theta) = \operatorname{argmax}_{\theta} \sum_{i=1}^N \log p(x_i | \theta)$$

Why can we do this?

- * $p(x_i | \theta) > 0$

- * $\log$ is monotonically increasing

Maximum Likelihood Estimation - The log trick



Example - MLE for the Gaussian Distribution

We consider i.i.d Gaussian random variables $\mathcal{D} = \{x_1, x_2, \dots, x_N\}$

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

• Likelihood: $p(\mathcal{D} \mid \mu, \sigma^2) = \prod_{i=1}^N \mathcal{N}(x_i \mid \mu, \sigma^2) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_i-\mu)^2}$

$(e^{a+b} = e^a e^b)$

$$= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} e^{-\sum_{i=1}^N \frac{1}{2\sigma^2}(x_i-\mu)^2}$$

• Log likelihood: $\log p(\mathcal{D} \mid \mu, \sigma^2) =$

$$= \log(1) - \frac{N}{2} \log(2\pi\sigma^2) - \sum_{i=1}^N \frac{1}{2\sigma^2}(x_i-\mu)^2$$

$$(\log(ab) = \log(a) + \log(b))$$

$$\log(a^N) = N \log(a)$$

Example - MLE for the Gaussian Distribution

$$\text{MLE of } \mu: \hat{\mu}_{ML} = \underset{\mu}{\operatorname{argmax}} P(D | \mu, \sigma^2) = \underset{\mu}{\operatorname{argmax}} \log P(D | \mu, \sigma^2)$$

$$\frac{\partial}{\partial \mu} \log P(\underline{x} | \mu, \sigma^2) = \frac{\partial}{\partial \mu} \left(-\frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2 \right)$$

$$= -\frac{1}{2\sigma^2} \sum_i \frac{\partial}{\partial \mu} (x_i - \mu)^2$$

$$= -\frac{1}{2\sigma^2} \sum_i 2(x_i - \mu) \cdot (-1) = \frac{\sum_i x_i - N\mu}{\sigma^2}$$

$$\frac{\partial}{\partial \mu} \log P(\underline{x} | \mu, \sigma^2) = 0 \Leftrightarrow \sum_i x_i = N\mu \Leftrightarrow \mu = \boxed{\frac{\sum x_i}{N} = \hat{\mu}_{ML}}$$

$$\left(\frac{\partial^2}{\partial \mu^2} = -\frac{N}{\sigma^2} < 0 \forall \mu \Rightarrow \log \text{llh is concave} \right) \Rightarrow \text{the optimal point is indeed the global max.}^{18}$$

Example - MLE for the Gaussian Distribution

Considering $\mu = \hat{\mu}_{ML}$, let's compute the MLE of σ^2 :

$$\log l(\mu) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

$$\frac{\partial \log p(x | \mu_{ML}, \sigma^2)}{\partial \sigma^2} = -\frac{N}{2} \cdot \frac{1}{2\pi\sigma^2} \cdot 2\pi + \frac{1}{2\sigma^4} \sum (x_i - \mu_{ML})^2$$

$$= \frac{N\sigma^2}{2\sigma^4} + \frac{\sum (x_i - \mu_{ML})^2}{2\sigma^4} \stackrel{!}{=} 0 \Leftrightarrow \sigma^2 = \left(\frac{\sum (x_i - \mu_{ML})^2}{N} = \hat{\sigma}_{ML}^2 \right)$$

↑ biased sample variance!

The sample variance is a biased estimator!

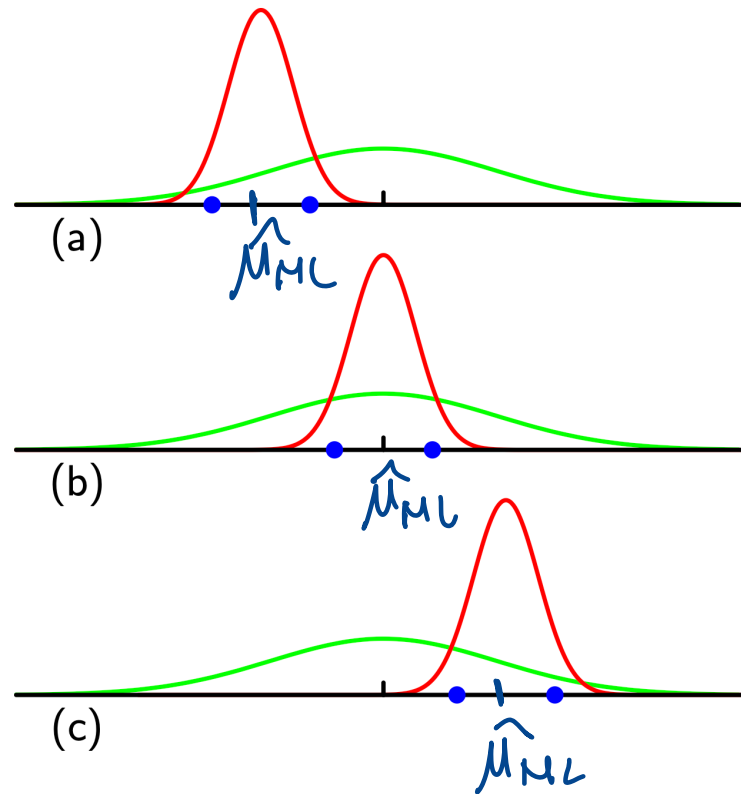


Figure: Gaussian distributions obtained by fitting to three data sets (red curves), each consisting of two data points shown in blue. The green curve shows the true Gaussian distribution from which the data is generated (Bishop 1.15)

$\hat{\sigma}_{ML}^2$ underestimates σ_{ML}^2 !!!

Maximum a Posteriori Estimation

The Posterior distribution

Suppose we have some prior information about a parameter θ , encoded as a density function $p(\theta)$.

We also have data $\mathcal{D}$ with likelihood $p(\mathcal{D} \mid \theta)$.

We can update the prior distribution with the data information encoded in the likelihood, obtaining the **posterior distribution**:

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})}$$

The Posterior distribution

$$p(\theta | \mathcal{D}) = \frac{p(\mathcal{D} | \theta)p(\theta)}{p(\mathcal{D})}$$

- Posterior: $p(\theta | \mathcal{D})$
- Likelihood: $p(\mathcal{D} | \theta)$
- Prior: $p(\theta)$
- Evidence: $p(\mathcal{D}) = \int_{\theta} p(\mathcal{D} | \theta)p(\theta)d\theta$ (sum + product rules)
intractable esp. if the model is very big

Maximum a Posteriori Estimation

Setup: N independent samples, identically distributed according to $p(x | \theta)$. We have prior knowledge about θ , encoded as a density function $p(\theta)$.

Goal: find the value of the parameter θ that maximizes the posterior

$$\hat{\theta}_{MAP} = \operatorname{argmax}_{\theta} p(\theta | \mathcal{D}) = \operatorname{argmax}_{\theta} \frac{p(\mathcal{D} | \theta)p(\theta)}{p(\mathcal{D})}$$

Since $p(\mathcal{D})$ does not depend on θ , this simplifies to:

$$\hat{\theta}_{MAP} = \operatorname{argmax}_{\theta} p(\mathcal{D} | \theta) p(\theta) = \operatorname{argmax}_{\theta} \log p(\mathcal{D} | \theta) p(\theta)$$

Example - MAP for the Gaussian with Gaussian Prior

$$x_i \sim \mathcal{N}(\mu, \sigma^2)$$

We consider i.i.d Gaussian random variables $\mathcal{D} = \{x_1, x_2, \dots, x_N\}$.

- The mean of the gaussian has a prior distribution $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$
- We consider the variance to be fixed.

We want to compute the Maximum A Posteriori estimate for μ :

- Likelihood: $p(\mathcal{D} | \mu) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} e^{-\sum_i \frac{1}{2\sigma^2} (x_i - \mu)^2}$

- Posterior: $p(\mu | \mathcal{D}) \propto \underset{\substack{\uparrow \\ \text{proportional} \\ \text{to}}}{p(\mathcal{D} | \mu)}} p(\mathcal{D} | \mu) p(\mu) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} e^{-\sum_i \frac{1}{2\sigma^2} (x_i - \mu)^2} \cdot \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{1}{2\sigma_0^2} (\mu - \mu_0)^2}$

Example - MAP for the Gaussian with Gaussian Prior

The optimization problem to solve is:

$$\hat{\mu}_{\text{MAP}} = \arg \max \log(p(D|\mu)p(\mu))$$

$$\begin{aligned} \log p(D|\mu)p(\mu) = & -\frac{N}{2} \log(2\pi\sigma^2) - \sum \frac{1}{2\sigma^2} (x_i - \mu)^2 - \frac{1}{2} \log(2\pi\sigma_0^2) \\ & - \frac{1}{2\sigma_0^2} (\mu - \mu_0)^2 \end{aligned}$$

$$\frac{d}{d\mu} \log \text{posterior} = +\frac{1}{\sigma^2} \sum (x_i - \mu) - \frac{1}{\sigma_0^2} (\mu - \mu_0) \stackrel{!}{=} 0$$

$$\Leftrightarrow \frac{+\sum x_i - \mu N}{\sigma^2} = \frac{\mu - \mu_0}{\sigma_0^2} \Leftrightarrow \sigma_0^2 (+\sum x_i - \mu N) = \sigma^2 (\mu - \mu_0)$$

$$\Leftrightarrow +\sigma_0^2 \sum x_i - \mu N \sigma_0^2 = \mu \sigma^2 - \sigma^2 \mu_0$$

$$\Leftrightarrow \mu = \frac{\sigma_0^2 \sum x_i + \sigma^2 \mu_0}{N \sigma_0^2 + \sigma^2} = \hat{\mu}_{\text{MAP}}$$

Linear interpolation between
prior mean and sample mean



Bayesian Inference

MAP Estimation vs Bayesian Inference

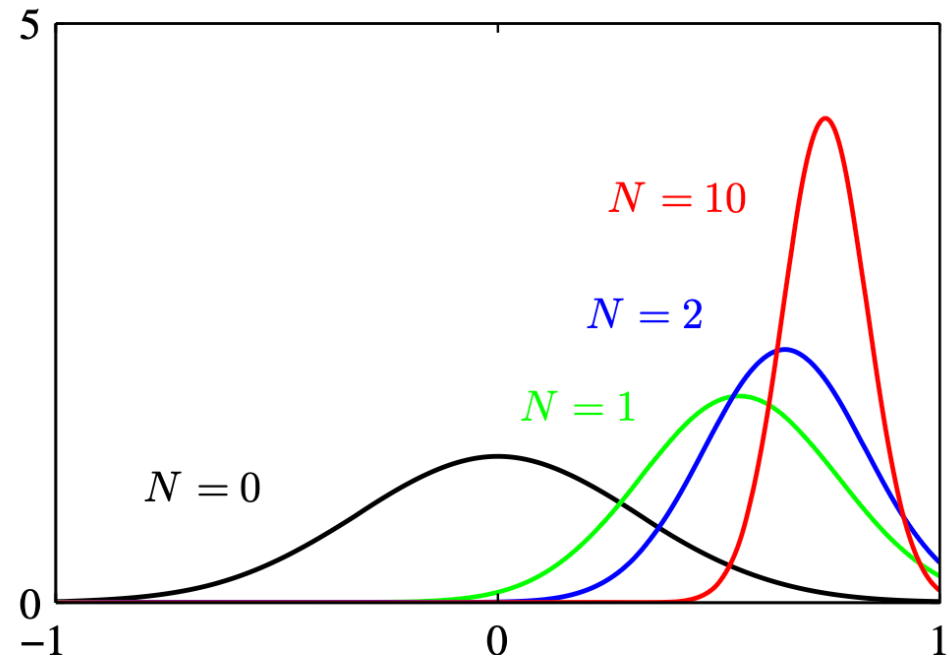
The MAP estimation method returns a single value as estimate of the parameter θ . However, we might also be interested in the full posterior distribution

$$p(\theta | \mathcal{D}) = \frac{p(\mathcal{D} | \theta)p(\theta)}{p(\mathcal{D})}$$

Example - MAP for the Gaussian with Gaussian Prior

Figure 2.12 Illustration of Bayesian inference for the mean μ of a Gaussian distribution, in which the variance is assumed to be known. The curves show the prior distribution over μ (the curve labelled $N = 0$), which in this case is itself Gaussian, along with the posterior distribution given by (2.140) for increasing numbers N of data points. The data points are generated from a Gaussian of mean 0.8 and variance 0.1, and the prior is chosen to have mean 0. In both the prior and the likelihood function, the variance is set to the true value.

(Source: Bishop)



Conjugate distributions

We say that a distribution $p(\theta)$ is a **conjugate prior** for a likelihood function $p(x \mid \theta)$ iff the posterior distribution $p(\theta \mid x)$ is in the same probability distribution family as the prior

Examples of conjugate distributions:

Likelihood	Conjugate prior
Gaussian	Gaussian
Bernoulli	Beta
Poisson	Gamma
Uniform	Pareto

...

Recap

	Frequentist approach	Bayesian approach
Point estimate	<i>MLE</i> $\hat{\theta}_{ML} = \operatorname{argmax}_{\theta} p(\mathcal{D} \mid \theta)$	<i>MAP</i> $\hat{\theta}_{MAP} = \operatorname{argmax}_{\theta} p(\theta \mid \mathcal{D})$
Distributional inference		<i>Bayesian inference</i> $p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})}$

That was it!

HW set 1 is out

Practical sessions start today and tomorrow!

Lab 1 is out – Deadline next Monday

Next Lecture: Linear Regression

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